Laser Thermal Beamed Energy Propulsion Feasibility Study

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The feasibility of laser thermal beamed energy propulsion was explored for small payload mass (40 - 400 kg) single-stage-to-orbit vehicles. Families of point designs were produced and analyzed using the trajectory optimization software ASTOS and in-house code containing existing mathematical models of atmospheric effects on the beam, the heat exchanger thermodynamics, and rocket propulsion mechanics. For each payload mass, the maximum payload fractions we found were near 3%, the absorbed power required for each kilogram of payload mass was near 2 MW/kg, and the power transmission efficiency (from ground to craft) was roughly 70%. This is compared to an existing single-stage-to-orbit chemical launch vehicle which achieves a 1% payload fraction.

I. Introduction

In a conventional launch vehicle, fuel reacts with an oxidizer and the resulting heated products expand from the nozzle to provide thrust. In laser thermal beamed energy propulsion (BEP), the energy is instead beamed from high-power laser groundstations (GSs) to a heat exchanger (HX) on the launch vehicle, where it is absorbed by the propellant. No longer constrained by the energy density of the fuel and oxidizer, this scheme can theoretically achieve greater propellant velocities, and therefore higher payload fractions.

In the BEP system studied here, the GSs are modular arrays of 1064 nm lasers, the HX is made of silicon carbide (SiC) with embedded microchannels through which liquid hydrogen (LH2) propellant flows. The craft is single-stage-to-orbit (SSTO), and we compare our craft performance to an existing chemical SSTO rocket design: the Delta Clipper [1].

A. Theoretical performance of thermal BEP

The aspect that limits the performance of a laser thermal BEP craft is the finite melting temperature of the HX material, rather than the finite energy stored in chemical bonds. SiC has a melting point of 2730 K [2], and let us assume that our SiC HX heats the LH2 propellant temperature T from $T_{\text{vap}} \approx 20$ K to 2500 K. With specific heat capacity c_{p}

This manuscript (paper number AIAA 2024-1810) was presented at the 2024 AIAA SciTech conference in Orlando, Florida.
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values at 1 atm and specific vaporization energy of $\frac{E_{\text{vap}}}{m} \approx 0.45\,\text{MJ/kg}$ taken from [3], the energy density is:*

$$\frac{E_{\text{BEP}}}{m} = \frac{E_{\text{vap}}}{m} + \int_{20}^{2500\,\text{K}} c_{\text{p}}\,\text{d}T \approx 39.1\,\tfrac{\text{MJ}}{\text{kg}}. \tag{1}$$

The payload fraction, PF, is the performance metric we focus on in this paper: the ratio m_{p}/m_0, where m_{p} is the payload mass and m_0 is the total mass of the rocket before takeoff. Using the Tsiolkovsky rocket equation [4] this can be expressed in terms of v_{LEO}, the propellant exit velocity v_{e}, and the dry mass m_{dry} (which includes m_{p}):

$$\text{PF} = \frac{m_{\text{p}}}{m_0} = \frac{m_{\text{p}}}{m_{\text{dry}}} \exp\left(\frac{-v_{\text{LEO}}}{v_{\text{e}}}\right) = \frac{m_{\text{p}}}{m_{\text{dry}}} \exp\left(-v_{\text{LEO}}\sqrt{\frac{m}{2E}}\right). \tag{2}$$

This assumes that all of the available thermal energy in the propellant is converted into kinetic energy of the products. When the same assumption is made for a conventional rocket using LH2 and LOx, the theoretical maximum exhaust velocity is $v_{\text{e,CHEM}} \approx 5.22\,\text{km/s}$ [5]. We can evaluate Eq. (2) for the case of chemical and BEP, so that the two can be compared:

$$\frac{\text{PF}_{\text{BEP}}}{\text{PF}_{\text{chem}}} = \exp\left(-v_{\text{LEO}}\left(\sqrt{\frac{m}{2E_{\text{BEP}}}} - \frac{1}{v_{\text{e,CHEM}}}\right)\right) \approx 1.87. \tag{3}$$

Here we used a typical value $v_{\text{LEO}} = 8\,\text{km/s}$. This back-of-the-envelope-calculation suggests that thermal BEP (using LH2 and a SiC HX) should achieve almost twice the payload fraction to LEO as a chemical rocket (using LH2 and LOx). This is a very approximate calculation - drag losses, gravity losses, staging, and other considerations have not been included. The main objective of this paper is to refine this value with a more detailed model.

B. Notable past work in BEP

In the early fifties before the laser existed, Sänger envisioned a *photonenrakete*; a rocket propelled by the momentum imparted by photons reflecting from its aft surface [6]. Two current projects involving light-driven crafts travelling beyond the Solar System are Parkin's *Breakthrough Starshot* scheme [7] and Lubin's *DEEP-IN* scheme [8]. These schemes require no onboard propellant but are more suited to in-space propulsion than to launch: a simple calculation shows that a laser (of any frequency) reflecting from a surface provides only $2/c \approx 6.67\,\text{N}$ of force per *gigawatt* of laser power (where c is the speed of light). Converting beamed energy into electrical energy (via photovoltaics) to then run electric thrusters has also been explored for in-space propulsion [9, 10].

For launch, proposed BEP schemes require heating onboard reaction mass. One strategy is to use pulsed lasers to ablate solid mass from the craft, as was studied by Myrabo with his *Lightcraft*† [12]. While in the atmosphere, the pulses were focused near the underside of the craft to heat the air for an airbreathing propulsion mode. See [13] for a

*The high specific heat capacity and low vaporization energy of Hydrogen is the reason we choose this as our propellant.

†One of his Lightcraft holds the record for the highest altitude achieved by a BEP craft: 71 m, set in the year 2000. See [11] for footage.

review of other BEP schemes relying on ablation.

Kantrowitz first proposed BEP for launch by heating onboard propellant [14] with ground-based lasers. Further work by Kare popularized the advantages of using modular non-coherent arrays of continuous lasers[15–18]. Using a continuous microwave beam to heat onboard propellant has also been studied [19]. One aim of our work is to build on Kare's analysis of the power requirements.

C. Intention of this work

NASA's Beamed Energy Propulsion study [20] contains a design reference mission for Kare's laser thermal scheme with modular non-coherent laser arrays. In their study, the goal was to formulate a single self-consistent scheme, or *point design*[‡], rather than find the optimal design in what is a very large parameter space. They summarize this goal:

> Finally, the intention of this study was to work within the time and resources allotted to determine the feasibility of BEP. This involved achieving a single closed-design solution for each DRM [Design Reference Mission], which is, in all probability, is [sic] not the optimum solution. As such, proponents of the investigated concepts may not agree with the findings and the study final design solutions, citing that better performance could be achieved with additional design effort.

The laser thermal craft design reference mission in [20] achieves a payload fraction of around 1.48%. In our study we sought to improve on this by generating mutiple point designs, and identifying the one that maximizes the payload fraction. We found that the most influential parameters (e.g., propellant mass flow rate and power absorbed by the propellant) could be expressed in terms of the HX area A, so we chose A as our optimization variable. Our results are in Section IV. The calculations for determining the paylaod fraction are in Section II and the calculations for determining the laser power required are in Section III.

II. Determining the payload fraction

The procedure to create a point design began by selecting a payload mass m_p and a HX area A. Next we chose a mass flow rate $\dot{m}$ and a power entering the propellant P_{H_2} which constrain the surface temperature of hottest regions of the HX (the ends of the microchannels) to $T = 2500\,\mathrm{K}$, and the initial acceleration of the craft to $>12\,\mathrm{m/s^2}$. The trajectory optimization software `ASTOS` [21] was then used to determine the payload fraction achieved.

[‡]It is called a *point design* because if we imagine a high-dimensionality mathematical space with an axis for every design parameter of the craft, a single design will occupy a *point* in this space.

A. Heat exchanger

The HX is a SiC ellipse of area A, with microchannels embedded for the propellant to flow through, separated from neighboring microchannels by walls we refer to as *fins*. LH2 enters a microchannel, and absorbs heat conductively through the HX face, heating it to near 2500 K before it expands through a nozzle. For each time step and for each area element of the HX above a microchannel, the power $\mathrm{d}P_{\mathrm{HX}}$ falling on the HX is apportioned as either power entering the fuel $\mathrm{d}P_{\mathrm{H_2}}$, power convectively lost through aerodynamic interactions $\mathrm{d}P_{\mathrm{aero}}$, or power reradiated $\mathrm{d}P_{\mathrm{radi}}$:[§]

$$\mathrm{d}P_{\mathrm{HX}} = \pm\mathrm{d}P_{\mathrm{aero}} + \mathrm{d}P_{\mathrm{radi}} + \mathrm{d}P_{\mathrm{H_2}}. \tag{4}$$

For clarity, calculation of $\mathrm{d}P_{\mathrm{aero}}$ and the reradiation losses $\mathrm{d}P_{\mathrm{radi}}$ will be left for Subsection III.C, where the other power loss mechanisms are treated. Here we are interested in finding how the HX surface temperature T is related to the propellant temperature $T_{\mathrm{H_2}}$. The temperature dependence of the thermodynamic properties of SiC were from [2], and its emissivity from [22]. The temperature dependence of the thermodynamic properties of the hydrogen propellant (the thermal conductivity $\kappa_{\mathrm{H_2}}$, specific heat capacity at constant pressure c_{p}, specific heat ratio γ, Reynold's number Re, and Prandtl number Pr) were found using the `C++` library `CoolProp` [3]. The dimensions of the HX microchannels were chosen by making slight alterations to the choices in Los Alamos' study of a nuclear thermal rocket engine [23].

The temperature drop across the HX face ($T - T_{\mathrm{H_2}}$) was calculated by forming a thermal resistance network, from which we could find the equivalent resistance of the face and the fin. The thermal equivalent of Ohm's Law is:

$$\left(T - T_{\mathrm{H_2}}\right) = \mathrm{d}P_{\mathrm{H_2}}\mathscr{R}_{\mathrm{system}}. \tag{5}$$

To find the equivalent resistance $\mathscr{R}_{\mathrm{system}}$, we added the fin-to-fluid resistance $\mathscr{R}_{\mathrm{fin-H_2}}$ in parallel to the face-to-fluid resistance $\mathscr{R}_{\mathrm{face-H_2}}$, then added the face resistance $\mathscr{R}_{\mathrm{face}}$ in series:

$$\mathscr{R}_{\mathrm{system}} = \mathscr{R}_{\mathrm{face}} + \frac{\mathscr{R}_{\mathrm{fin-H2}}\mathscr{R}_{\mathrm{face}}}{\mathscr{R}_{\mathrm{fin-H_2}} + \mathscr{R}_{\mathrm{face-H_2}}}. \tag{6}$$

The resistances are:

$$\mathscr{R}_{\mathrm{face}} = \frac{d_{\mathrm{f}}}{\kappa \mathrm{d}A}, \qquad \mathscr{R}_{\mathrm{face-H_2}} = \frac{1}{H\mathrm{d}A}, \qquad \mathscr{R}_{\mathrm{fin-H_2}} = \frac{1}{\eta_{\mathrm{fin}} H \mathrm{d}A_{\mathrm{fin}}}. \tag{7}$$

Here, H is the heat transfer coefficient between the fluid and the SiC (assumed to be the same value for transfer into the fin or the face), η_{fin} is the efficiency of the fin, and $\mathrm{d}A_{\mathrm{fin}}$ is the area of the fin element. η_{fin} can be found if we assume the

[§]Note that this power falling on the HX is whatever remains once the power lost in transmission from the GS has been subtracted. This is the subject of Section III.

fin has an adiabatic end [24]:

$$\eta_{\text{fin}} = \frac{\tanh\left(\sqrt{\frac{HcL_{\text{fin}}^2}{\kappa \mathrm{d}A_{\text{fin}}}}\right)}{\sqrt{\frac{HcL_{\text{fin}}^2}{\kappa \mathrm{d}A_{\text{fin}}}}}. \tag{8}$$

Here, c is the perimeter of the fin, and L_{fin} is the length of the fin (which is the channel height minus the channel width for our pill-shaped channel). The heat transfer coefficient is:

$$H = \frac{\kappa_{\text{H}_2}\text{Nu}}{D_{\text{H}}}, \qquad D_{\text{H}} \equiv \frac{4A_{\text{H}}}{L_{\text{H}}}. \tag{9}$$

κ_{H_2} is the thermal conductivity of the fluid at that point, D_{H} is the hydraulic diameter, A_{H} is the area of the channel element, L_{H} is the perimeter of the wetted surface, and Nu is the local Nusselt number at that position. The Nusselt number differs for laminar and turbulent flow; while the fluid flow was laminar, the heat transfer coefficient between the fluid and the channel walls was found by following [25], and while the flow was turbulent the Dittus-Boelter correlation was used [26]:

$$\text{Nu}_{\text{turbulent}} = 0.023\text{Re}^{0.8}\text{Pr}^{0.4}\left(1 + \left(\frac{D_{\text{H}}}{x}\right)^{0.7}\right), \qquad \text{Nu}_{\text{laminar}} = 4.36 + \frac{0.036\left(\frac{\text{RePr}}{x/D_{\text{H}}}\right)}{1 + 0.0011\left(\frac{\text{RePr}}{x/D_{\text{H}}}\right)}. \tag{10}$$

Here, Re is the Reynold's number, Pr is the Prandtl number, and x is the coordinate directed along the HX microchannel length. With the heat transfer coefficient and resistances now specified, the temperature difference between the HX surface and the fluid can be calculated using Eq. (5). To determine the HX surface temperature at each element along a microchannel, we need a relationship between the fluid temperature and the power absorbed:

$$\frac{\mathrm{d}P_{\text{H}_2}}{\mathrm{d}x} = \frac{\dot{m}}{N} c_{\text{p}} \frac{\mathrm{d}T_{\text{H}_2}}{\mathrm{d}x}. \tag{11}$$

That is, the fluid temperature increases at each length element in accordance with the power absorbed at that step. Given the aerocooling and reradiation losses (we will find these in the next section), we can now solve the power balance Eq. (4) at each length element along the HX microchannel, for each time step, and find the propellant temperature T_{H_2} as it exits the HX.

B. Trajectory simulations

Armed with T_{H_2} and $\dot{m}$, simulations in `ASTOS` could be performed. `ASTOS` requires an engine model: a function relating the thrust F to the mass flow rate $\dot{m}$:

$$F = \dot{m} v_{\text{e}} + (p_{\text{e}} - p_0) A_{\text{t}}\text{ER}. \tag{12}$$

Here, p_e is the exit pressure, p_0 is the ambient pressure, ER is the nozzle expansion ratio, and A_t is the nozzle exit area. A_t was found this by rearranging the choked flow equation [27]:

$$A_t = \dot{m}\left[\gamma\rho_{H_2}p_{H_2}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}\right]^{-1/2}. \tag{13}$$

Here, ρ_{H_2} is the propellant density, γ is the specific heat ratio of the propellant, and p_{H_2} is the chamber pressure. p_{H_2} was found by setting the tank pressure to 10 MPa and then tracking the total pressure drop through the HX microchannels using the Darcy-Weisbach equation [28]. Following Stark's investigation [29], the exit area $A_e = A_t ER$ was chosen such that the resulting Mach number Ma gave a nozzle exit pressure that was equal to the critical value at which flow separation occurs (for sea level operation):

$$p_e = p_{H_2}\left(1+\frac{\gamma-1}{2}\mathrm{Ma}^2\right)^{\frac{-\gamma}{\gamma-1}} \geq \frac{\pi p_0}{3\mathrm{Ma}}, \tag{14}$$

where the Mach number is related to the exit area by an isentropic flow equation [30]:

$$A_e = \frac{A_t}{\mathrm{Ma}}\left(\frac{2}{\gamma+1}+\frac{\gamma-1}{\gamma+1}\mathrm{Ma}^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}. \tag{15}$$

The propellant velocity and temperature at the nozzle exit can also be found with isentropic flow equations [30]:

$$v_e = \mathrm{Ma}\sqrt{\gamma\frac{R}{M}T_e}, \qquad T_e = T_{H_2}\left(1+\frac{\gamma-1}{2}\mathrm{Ma}^2\right)^{-1}. \tag{16}$$

Here, R is the universal gas constant, M is the molecular mass of the propellant, and T_e is the propellant temperature at the nozzle exit.

With the thrust now specified by Eq. (12), `ASTOS` determined the craft trajectory that reached our specified orbit with the minimum propellant mass (i.e., maximum payload fraction). The `ASTOS` simulations had the following parameters and constraints: a single-stage craft with two flight phases (a constant pitch phase, then a phase with freedom to vary the pitch and yaw as needed), a fixed payload mass m_p, a fixed structural mass of $4m_p$, a fixed HX mass (calculated from the HX area), targeting a sun-synchronous orbit, with a 97.4° final inclination, originating from Wallops Flight Facility, with a 180 km perigee and 500 km apogee altitude (circularisation burn ignored), and freedom to reduce $\dot{m}$ (although `ASTOS` chose to have this maximized for the entire flight). The maximum acceleration, maximum dynamic pressure, and maximum bending moment were constrained to be less than 80 m/s^2, 25 kPa, and 60 kNm, respectively. Trajectory results from `ASTOS` are shown in Figure 1, for our chosen 400 kg point design which in detailed in Section IV.

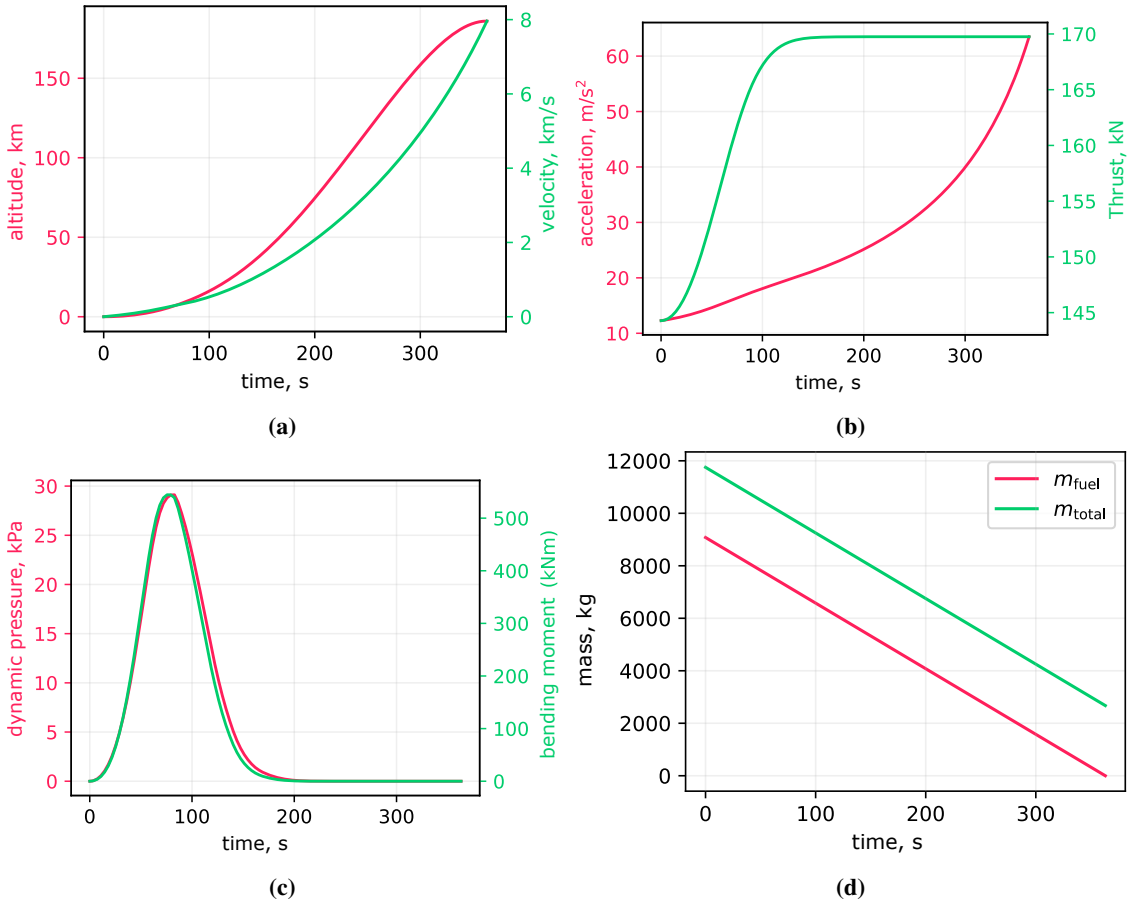


Fig. 1 Trajectory information from ASTOS for the 400 kg payload point design.

III. Determining the laser power required

The previous section related the beam power that struck the HX to the power absorbed by the propellant. In this section we find how much laser power needs to be transmitted from each GS in order for the requisite power to reach the HX, and ultimately reach the propellant. To this end, the power loss mechanisms are described in this section. The amount of power loss ultimately determines the number of groundstations required, and their spacing.

Following [16], each GS consists of many identical laser modules. We have assumed that each module is a 300 kW laser, outfitted with telescopic optics to expand and collimate the beam before it leaves the module. At each point in time, whichever GS can supply the requisite power to the propellant with the least losses is designated the active GS. The beam power P_{GS} leaving a GS, required such that the propellant receives P_{H_2}, is

$$P_{\text{GS}} = P_{\text{att}} + P_{\text{spill}} + P_{\text{refl}} + P_{\text{aero}} + P_{\text{radi}} + P_{\text{H}_2} = \frac{P_{\text{H}_2}}{\eta_{\text{att}}\,\eta_{\text{spill}}\,\eta_{\text{refl}}\,\eta_{\text{aero}}\,\eta_{\text{radi}}}. \tag{17}$$

Here, P is the power consumed by each mechanism, and η the corresponding efficiency[¶]. The subscripts correspond to the following power loss mechanisms, in order: atmospheric attenuation, spillover (from the beam area exceeding the visible HX area), power reflected from the HX, the HX losing heat to the atmosphere through aerodynamic mechanisms, and finally the HX losing heat through blackbody radiation.

A. Attenuation

Total beam attenuation was calculated by integrating the elemental attenuation for each groundstation, for each timestep, for each length element along the active beam. We used the attenuation coefficients from [31] for molecular absorption and molecular scattering, in summertime, for 1064 nm light. Following [32] we have ignored the coefficients related to aerosol absorption and scattering, since at our power levels the aerosols will be quickly vaporized and cease to cause any attenuation. We label the sum of our two attenuation coefficients a_0 and assume that the attenuation decreases with altitude in proportion to the air density. We find the efficiency corresponding to atmospheric attenuation is:

$$\eta_{\text{att}} = \exp\left(\frac{z_{\text{s}} a_0}{\cos\theta} \mathrm{e}^{-z_{\text{GS}}/z_{\text{s}}} \left(\mathrm{e}^{-s\cos\theta/z_{\text{s}}} - 1\right)\right). \tag{18}$$

Here, z_{s} is the scale height of the atmosphere, z_{GS} is the altitude of the active GS, and s and θ are the range and zenith angle of the craft with respect to the GS.

B. Spillover

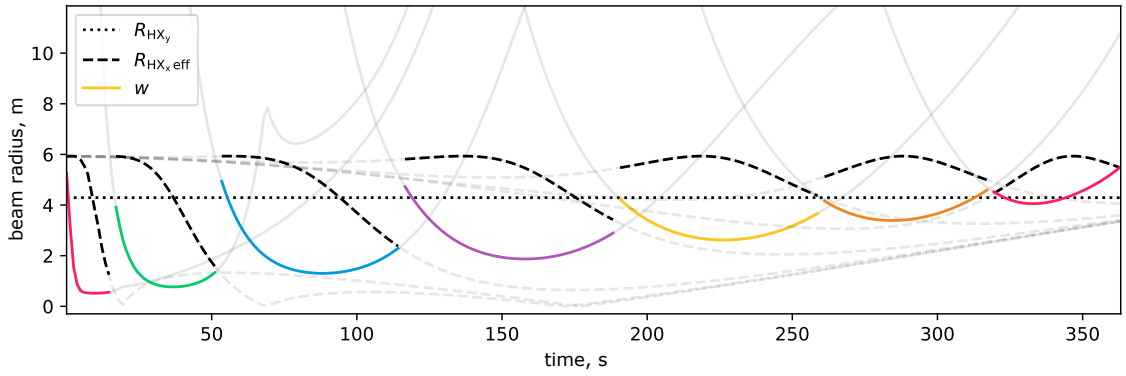
Spillover is the power lost due to the beam being larger than the visible HX area. The HX is mounted to the side of the craft, so if at any point the craft is oriented directly away from the groundstation then the beam will have zero *visibility* of the HX. Conversely if the craft is oriented to fly perpendicular to the beam, then the visibility is a maximum. For each point in time, and for each groundstation, the projected/effective/visible radius of the HX is:

$$R_{\text{HX}_\text{x}\,\text{eff}} = |\hat{\mathbf{s}} \times \hat{\mathbf{v}}|\, R_{\text{HX}_\text{x}}. \tag{19}$$

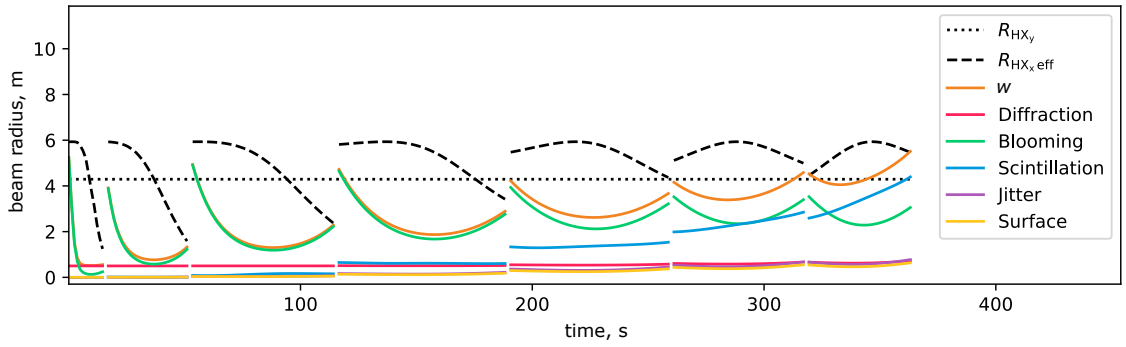
Here R_{HX_x} is the physical semi-major radius of the HX (oriented from the craft tail to nose). This equation is valid if we assume that the craft has zero angle of attack, and that the projection of the craft on the Earth's surface is always moving directly toward or away from the active groundstation.

To calculate the spillover, we integrate the beam over the visible area of an elliptical HX. Here, the x-direction is along the HX semi-major axis (tail to tip), and the y-direction is along the semi-minor axis. With beam radius w and

[¶]Recall from Eq. (4) that $P_{\text{HX}} = \pm P_{\text{aero}} + P_{\text{radi}} + P_{\text{H}_2}$ involves only two loss mechanisms. In this section we model the other mechanisms.



(a)



(b)

Fig. 2 (a) Beam radius during launch, for the 400 kg payload point design. (b) Breakdown of the beam broadening mechanisms that determine the beam radius.

effective HX area $A_{\text{eff}} = \pi R_{\text{HX}_y} R_{\text{HX}_x\,\text{eff}}$, the efficiency corresponding to spillover losses is:

$$\eta_{\text{spill}} = \frac{2}{\pi w^2} \iint_{A_{\text{eff}}} \exp\left(\frac{-2\left(x^2+y^2\right)}{w^2}\right) \mathrm{d}x\mathrm{d}y. \tag{20}$$

We next need to determine w. As the beam travels, it will broaden. We consider the following beam broadening mechanisms: diffraction, scintillation, thermal blooming, mechanical jitter, and optical surface figure errors. The total beam radius is shown in Fig. 2a, and this is broken into individual beam broadening mechanisms in Fig. 2b. In the plane transverse to the beam, each of these broadening mechanisms (in the absence of the others) would widen an infinitesimally narrow beam to one with a Gaussian intensity profile. To calculate the resultant beam width, a convolution integral can be used to combine the individual Gaussian functions. When Gaussian functions are convolved, the widths simply add in quadrature [33], giving the resulting beam radius at the craft:

$$w = \sqrt{w_{\text{diff}}^2 + w_{\text{scint}}^2 + w_{\text{bloom}}^2 + w_{\text{jitter}}^2 + w_{\text{surface}}^2}. \tag{21}$$

The craft performance would suffer if the beam radius was far smaller than the HX; the power should instead be spread evenly over the HX. Fortunately since our GS is modular, each individual laser module can be steered slightly differently to spread the power more evenly.

We assume our beam begins as a circular Gaussian of width w_0 and wavelength λ. From the well-known diffraction equation, the radius at range s is:

$$w_{\text{diff}} = w_0 \left[1 + \left(\frac{s}{s_{\text{R}}} \right)^2 \right], \qquad s_{\text{R}} \equiv \frac{\pi w_0^2}{2\lambda}. \tag{22}$$

Scintillation does not broaden a beam, strictly speaking. Scintillation is a fast and random deflection of the beam due to turbulence in the atmosphere. The turbulence brings cells of air with random temperatures (hence random densities, hence random refractive indices) through the beam path. Over a timescale longer than the average deflection frequency (the *Greenwood* frequency), an infinitesimally small beam *paints* a Gaussian distribution of half-width w_{scint}. The size of these deflections is given by the scintillation index C_n^2, also known as the *refractive index structure parameter* or the *turbulence strength*, which we derived using the Hufnagel-Valley model; see equation 13 in [34].

$$w_{\text{scint}} = \frac{1.45\lambda s}{\pi} r_0^{-5/6} w_0^{-1/6}, \qquad r_0 \equiv \left[0.423 \left(\frac{2\pi}{\lambda} \right)^2 \int_{z_{\text{GS}}}^{s\cos\theta + z_{\text{GS}}} \frac{C_n^2(s,\theta)}{\cos\theta} \mathrm{d}s \right]^{-3/5}. \tag{23}$$

Scintillation is a distortion of the beam due to random air temperature fluctuations in the beam path, whereas *thermal blooming* is a distortion of the beam due to the temperature rise of the air from being heated by the beam itself. Thermal blooming is asymmetric for slanted beams; since heat rises, the channel of heated air is taller than it is wide, which results in a slightly elliptical beam. However, in our power regime this ellipticity is small, so we approximated this beam component as circular, conservatively taking the larger radius (the y-oriented radius). The model we use is from [35]:[‖]

$$w_{\text{bloom}} = w_0 \alpha_1 s \sqrt{\left(1 + K_2 \Phi^2\right) + (1 - Q)\sqrt{\frac{\pi}{2}} \left(\frac{2}{\alpha_1 L} \right) \Phi} - \frac{2s}{k w_0}. \tag{24}$$

The ratio of energy deposited along the transverse axes Q, the thermal blooming critical power P_{T}, and the thin-screen phase shift Φ are also required for this calculation:

$$Q = \frac{P_{\text{T}}^2 + K_1 P_0^2 \frac{L\alpha_1}{4}}{2P_{\text{T}}^2 + (K_1 + K_2) P_0^2 \frac{L\alpha_1}{4}}, \qquad P_{\text{T}} = \frac{\sqrt{2}\pi\rho\bar{v}_{\text{wind eff}} C_{\text{p}}}{k^2 w_0 |\varepsilon_{\text{T}}| \alpha}, \qquad \Phi = \left| \frac{P_0}{P_{\text{T}}} \left(\frac{L}{2k w_0^2} \right) \right|. \tag{25}$$

[‖] There is more uncertainty in this beam broadening mechanism than in the others; [36] (by the same author) states that "analytical treatments appropriate for an engineering analysis of atmospheric propagation systems encountering thermal blooming have been lacking." One obvious problem is that in this model, the entire atmosphere is treated as an infinitesimally thin layer, or *phase screen*, which means that the model has no dependence on zenith angle θ. To account for this, we have modified the scale height to $z_{\text{s}}/\cos\theta$. With this modification a horizontal beam ($\theta = 90°$) gives an infinite beam radius, but since none of our point designs have horizontal beams, we decided it was a worthwhile modification. It is encouraging that with modular lasers the power of each laser module can be decreased (and the quantity increased) to lessen this thermal blooming.

Here, $K_1 = \frac{5}{3}$, $K_2 = \frac{16\pi}{9\sqrt{3}}$, $\alpha_1 = \frac{2}{kw_0^2}$, $L = z_s/\cos\theta$ is the scale height of the atmosphere, $k = \frac{2\pi}{\lambda}$ is the usual wavenumber, and ρ, C_p, α and ε_T are the atmospheric density, heat capacity, optical thickness, and rate of change of permittivity with temperature, respectively. $\bar{v}_{\text{wind eff}}$ is the effective wind speed, i.e., the ambient wind speed added to the slew rate of the beam, averaged along the beam path.

Finally, there will be beam broadening due to mechanical jitter of the beam module (e.g., from any coolant flow, the drive mechanism, etc.) and due to mechanical surface figure errors of the optics module. Following [16] we use the RMS errors of $\sigma = 1\,\mu$ radians for both errors, giving:

$$w_{\text{jitter}} = w_{\text{surface}} = \sqrt{8}\,\sigma s \tag{26}$$

C. Reflection, aerocooling, reradiation

At wavelength 1064 nm the reflectivity $\mathscr{R}$ of SiC is about 15% [22], with negligible dependence on temperature [22]. The efficiency associated with reflectivity is:

$$\eta_{\text{refl}} = 1 - \mathscr{R}. \tag{27}$$

For BEP we have to solve the same aeroheating problem as for conventional launch, except in this case the craft is cooled by the atmosphere rather than heated (as the HX is around 2500 K). If the power incident on the HX (once attenuation, spillover, and reflection losses have been subtracted) is $P_{\text{GS}}\eta_{\text{att}}\eta_{\text{spill}}\eta_{\text{refl}}$ then the efficiency associated with aerocooling can be found from Newton's law of cooling:

$$\eta_{\text{aero}} = 1 - \frac{H_{\text{aero}}A(T_{\text{rec}} - T)}{P_{\text{GS}}\eta_{\text{att}}\eta_{\text{spill}}\eta_{\text{refl}}}. \tag{28}$$

Here, the heat coefficient H_{aero} and recovery temperature T_{rec} are found using the Meador-Smart method [37], and A and T are the area and surface temperature of the HX. We assume the craft has no angle of attack.

The HX will radiate power according to the Stefan-Boltzmann law. If the power incident on the HX (once attenuation, spillover, reflection, and aerocooling losses have been subtracted) is $P_{\text{GS}}\eta_{\text{att}}\eta_{\text{spill}}\eta_{\text{refl}}\eta_{\text{aero}}$, then the efficiency associated with reradiation is:

$$\eta_{\text{radi}} = 1 - \frac{\varepsilon_{\text{HX}}\sigma_{\text{SB}}A\left(T^4 - T_{\text{ambient}}^4\right)}{P_{\text{GS}}\eta_{\text{att}}\eta_{\text{spill}}\eta_{\text{refl}}\eta_{\text{aero}}}. \tag{29}$$

Here, ε_{HX} is the emissivity of the HX material, σ_{SB} is the Stefan-Boltzmann constant, and T_{ambient} is the temperature of the Earth's surface below the craft.

D. Combined power loss and efficiency

For the 400 kg payload point design the placement of the GSs is shown in Fig. 3, and the power required by each during the launch is shown in Fig. 4. Note that there are seven GSs in this point design. For all of our point designs, we found that five to seven GSs were required, such that the power transmitted from the GSs would not become excessive. The power increases because as the craft recedes toward the horizon (from the perspective of the active GS), the beam must slant through more atmosphere which intensifies deleterious atmospheric effects, and the visible HX area reduces as the beam-to-HX angle of incidence increases.

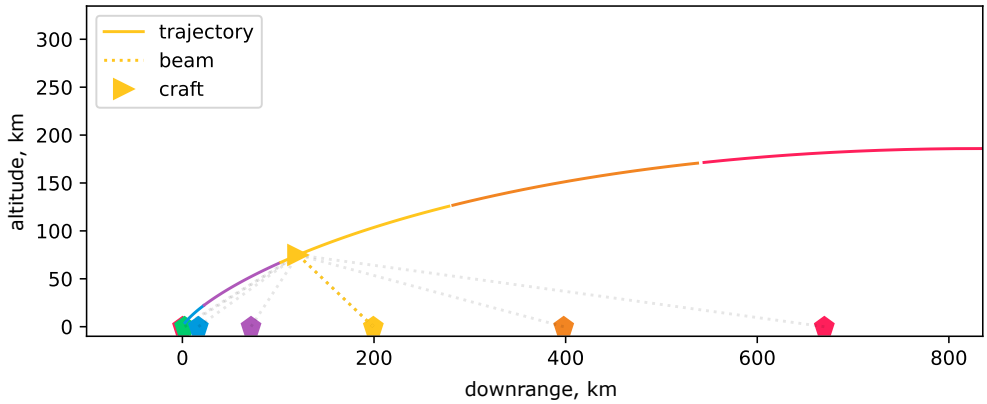


Fig. 3 Groundstation placement for the 400 kg payload point design. The craft is the yellow arrow, shown at its position 200 s after launch. The fifth groundstation (yellow pentagon) is powering the craft at this time, as it can do so at lower power output than the rest of the groundstations.

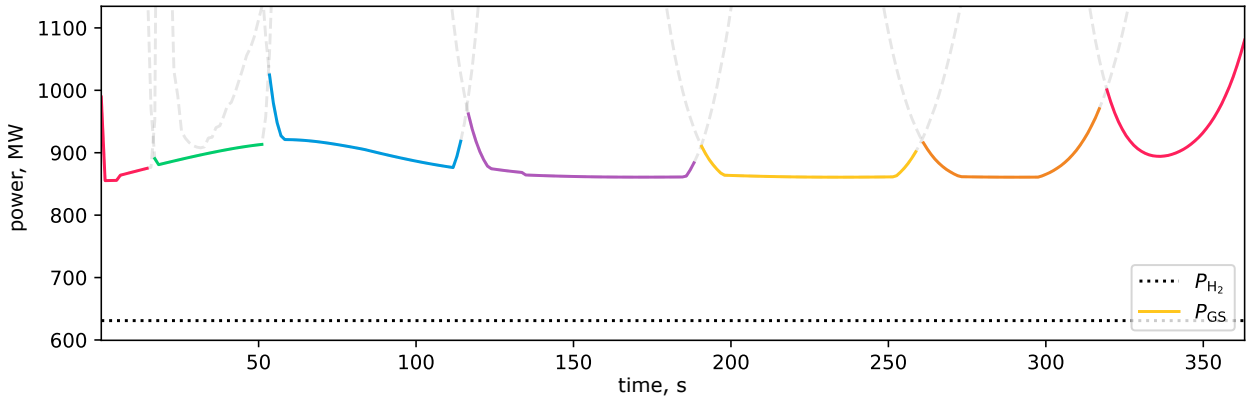


Fig. 4 Power output required to keep the HX at the maximum allowable temperature for the 400 kg payload point design.

A breakdown of Fig. 4 into each of these loss mechanisms is shown in Fig. 5, and their associated efficiencies are shown in Fig. 6.

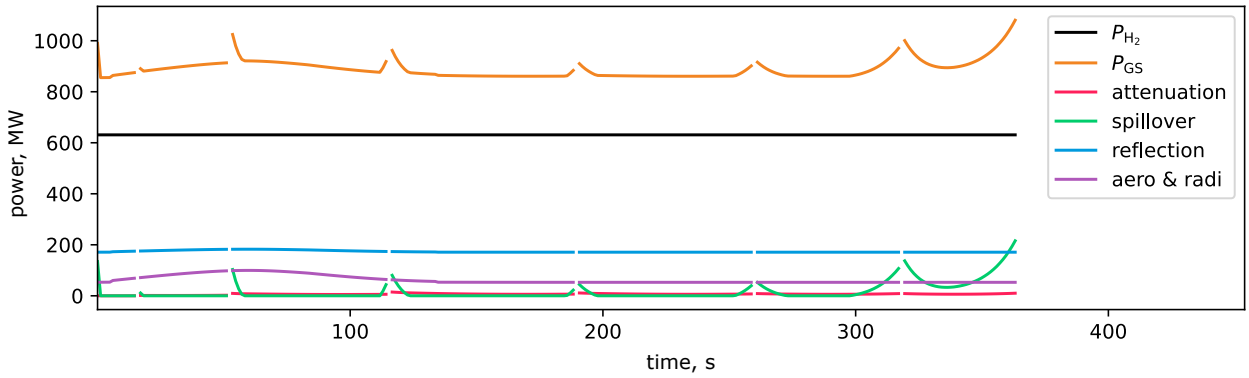


Fig. 5 Breakdown of the power loss mechanisms, for the 400 kg payload point design.

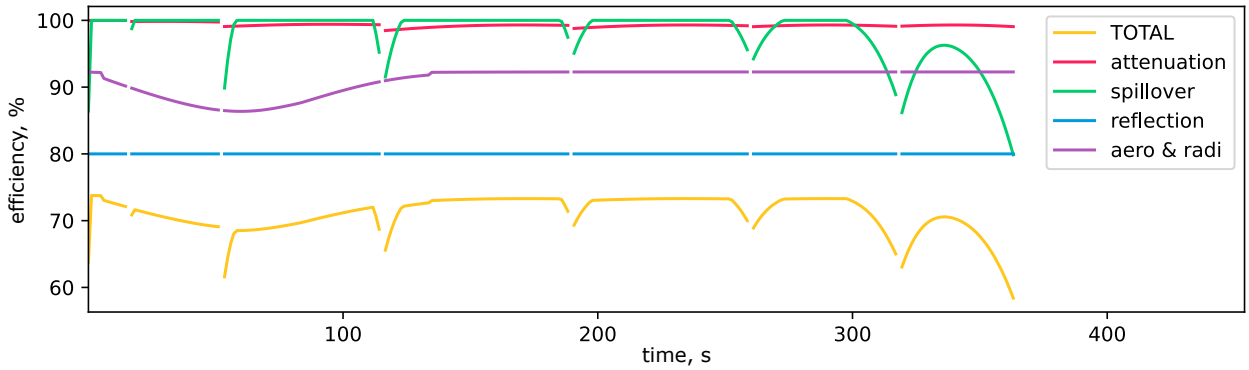
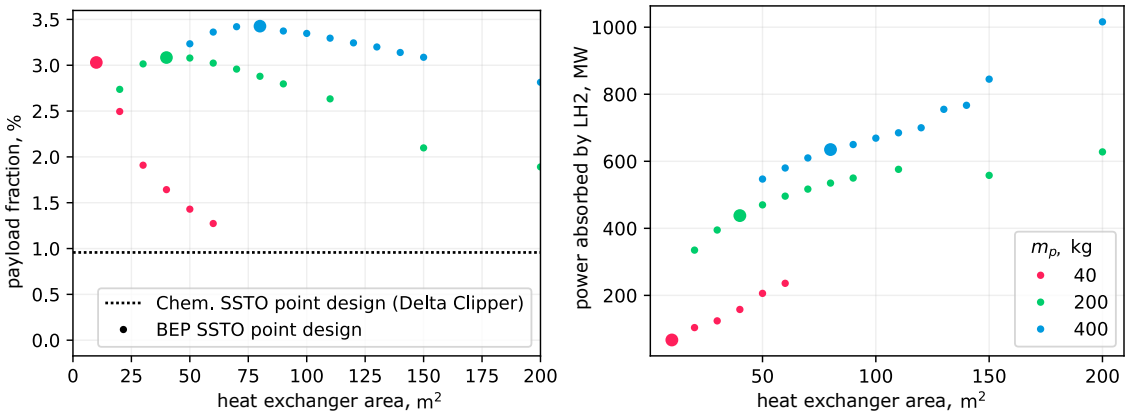


Fig. 6 Efficiencies associated with the power loss mechanisms, for the 400 kg payload point design.

IV. Point designs

We have explained how to combine HX calculations, `ASTOS` simulations, and the power loss calculations to produce a point design. Families of point designs were created for a range of HX areas and payload masses. Fig. 7a shows the payload fraction achieved in these designs, and Fig. 7b shows the power (transferred to the propellant) required. The chemical rocket design used in these plots for comparison is the Delta Clipper [1]; a reusable SSTO craft which supports a payload of around 4500 kg with an I_{SP} of 330 s at sea level and 425 s in vacuum, reaching a similar orbit (300 km by 300 km) as our point designs. The three point designs corresponding to the larger circular data points in Fig. 7a are detailed in Table 1. These are our point designs that are nearest to the maximum payload fraction.

We can understand qualitatively why an optimum exists by considering extreme values of A. For small A values, the greater beam intensity (power/area) incident on the HX results in a greater temperature drop across the face of the HX, giving a lower propellant temperature, lower velocity, and therefore a lower payload fraction. For large A values, the greater mass of the HX (and accompanying extra propellant mass) results in a lower payload fraction.



(a) (b)

Fig. 7 (a) The payload fraction achieved by point designs with different HX area and payload masses (m_p), as shown in the legend of 7b). The designs closest to an optimum are indicated by a larger data point. (b) The power (entering the propellant) required in each point design.

Table 1 Details of the three point designs corresponding to the larger circles in Fig. 7. Calculations for most of these quantities are given in Sections II and III.

CRAFT PARAMETER	●	●	●
Mass of payload (kg)	40	200	400
Mass of HX (kg)	84	337	674
Mass of other (kg)	160	800	1600
Mass of propellant (kg)	1123	5147	9077
Mass total (kg)	1407	6484	11751
Propellant pressure after HX (MPa)	9.99	9.97	9.98
Propellant temperature after HX (K)	1694	1442	1619
Power absorbed by propellant (MW)	67	438	631
Propellant mass flow rate (kg/s)	2.5	20.0	25.0
Propellant exit temperature (K)	549	465	524
Propellant exit pressure (kPa)	30	31	30
Propellant exit velocity (m/s)	7118	6509	6940
Nozzle expansion ratio	25.89	25.09	25.65
Nozzle throat area (m^2)	0.000988	0.007284	0.009658
Acceleration (initial) (m/s^2)	11.3	18.1	13.2
Thrust (sea level, vacuum) (kN)	(15.9, 18.6)	(117.2, 135.9)	(155.6, 181.0)
Specific impulse (sea level, vacuum) (s)	(649, 757)	(597, 693)	(634, 738)
Payload fraction (%)	2.8	3.1	3.4
Thrust-to-weight ratio	1.2	1.8	1.3
Mass of payload per P_{H_2} (kg/MW)	0.60	0.46	0.62

The HX design and calculation results for the 400 kg payload point design are shown in Table 2, and the GS placement and calculation results are in Table 3.

Table 2 Heat exchanger details for the 400 kg payload point design.

HX PARAMETER	Value
Mass of payload (kg)	400
Mass of HX (kg)	674
Surface temperature, hottest (K)	2362
Heat exchanger radii (x,y) (m)	(5.93, 4.29)
Power absorbed (MW)	593.15
Power reradiated (MW)	37.85
Reradiation peak wavelength (μm)	1.227
Energy lost to aerocooling (GJ)	3.02
Mass flow rate (kg/s)	25.00
Propellant temperature (initial, final) (K)	(34, 1619)
Propellant pressure (initial, final) (MPa)	(10.00, 9.98)
Channel shape	pill
Channel length (mm)	400
Channel height (mm)	2.1
Channel width (mm)	1.0
Channel fin width (mm)	0.3
Face thickness (mm)	1.0
Filling factor	0.9

Table 3 Groundstation placement and further details for the 400 kg payload point design. The bracketed values are for each station, ordered by proximity to the launch pad.

GROUNDSTATION PARAMETER	Value
Mass of payload (kg)	400
Beam power at module (kW)	300.0
Beam radius at module (m)	0.5
Beam wavelength (nm)	1064.0
GS altitudes (km)	(0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0)
GS downrange distances (km)	(0.1, 2.3, 16.6, 71.6, 199.3, 398.0, 670.0)
GS max slew rate (degrees/s)	(7.6, 2.6, 1.5, 1.4, 1.5, 1.7, 2.1)
GS max power output (MW)	(989.4, 957.5, 1024.5, 962.2, 912.9, 985.8, 1080.6)
GS time spent turned on (s)	(17.0, 36.4, 63.1, 74.1, 70.4, 58.3, 44.9)
Time craft passes over each GS (s)	(16.0, 50.0, 105.0, 170.0, 235.0, 290.0, 340.0)
Closest the craft comes to each GS (km)	(0.1, 2.0, 13.1, 45.6, 98.0, 148.6, 181.2)
Energy output of each GS (GJ)	(13.74, 31.68, 56.13, 63.28, 59.96, 50.26, 41.07)
Total energy output (GJ)	316.1
Number of 300 kW lasers	(3298, 3192, 3415, 3207, 3043, 3286, 3602)
Total number of 300 kW lasers	23043
Payload mass / total energy (kg/GJ)	1.27

V. Conclusion

We have described our feasibility study of laser thermal beamed energy propulsion for a single-stage-to-orbit light payload mass launch vehicle, with a focus on the payload fraction that the craft can achieve, the power required to be absorbed by the propellant, and the total power required at each groundstation. Existing models for beam broadening mechanisms, beam power loss mechanisms, heat exchanger thermodynamics, rocket propulsion mechanics, and trajectory design (`ASTOS`) were synthesized to form our study.

Such a scheme involves many free parameters, but we found we could express the most influential parameters in terms of the heat exchanger area, which allowed us to find the maximum payload fraction. We found, for each payload mass, the payload fraction maximum was near 3% (specifically 3.4% for the 400 kg payload point design that we discussed most). This is the first of our three salient results (see the larger dots in Fig. 7a).

Our second key result is that roughly 2 MW/kg of power absorbed in the propellant is required for each kilogram of payload mass (see the last line of Table 1).

Our third important result is that the power transmission efficiency is roughly 70%, and five to seven groundstations are needed to keep the required power level from becoming excessive as the heat exchanger visibility from the perspective of the groundstation diminishes (see the yellow line in Fig. 6).

With today's cost of high-power laser modules, we believe that a laser thermal BEP scheme is not worth the modest increase in payload fraction it promises.

Acknowledgements

We would like to thank Peter Beck for the opportunity to research this exciting topic, and we would also like to thank Adrian Pfeifle, Diego Pinto, Doug Sinclair, George Buchanan, Jonny Currie, Lachy Fitzpatrick, Marcel Scherrmann, Phil Dahm, and Will Snell for their help.

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